

Title: Renormalization as Survivability Bookkeeping

Status: Trailhead (STUB; required bridge for AoC constants + SM/QFT layer)

Goal

Recast renormalization, running couplings, and EFT cutoffs as survivability bookkeeping under $\zeta/\omega/\xi$ —so “scale dependence” is not a circular parameter chase, but an explicit record of how effective descriptions persist across constrained regimes.

Core Claim

RG flow is not merely a mathematical artifact of QFT regularization; it is the trace of constraint management: as resolution changes, ζ (bounds), ω (extension), and ξ (selection/noise) redistribute what remains coherent. “Running” encodes which interaction descriptions survive at each admissible scale window.

Must Include

- PF definition of scale: what changes when “you zoom in/out” (resolution, coarse-graining, collapse depth), and how that maps to $\zeta/\omega/\xi$ without assuming a fundamental background clock.
- Units Firewall compliance: identify which quantities are allowed to run (effective couplings, operator coefficients) vs which must remain invariant (dimensionless survivors / AoC invariants).
- EFT dictionary: translate standard EFT constructs (cutoff Λ , counterterms, operator expansion, decoupling) into PF language (bounds windows, stability domains, survivability pruning).
- Non-circularity proof sketch: show how PF can predict or constrain flow without fitting the same data it claims to explain.
- Discriminator: a concrete test comparing (a) PF-predicted flow family vs (b) purely empirical RG fit—include decision rule and what would falsify PF.

Outputs

- A compact PF $\leftrightarrow$ EFT/RG translation table (one-page minimum).
- A “running ledger” template: scale window $\rightarrow$ active effective operators $\rightarrow$ stability notes $\rightarrow$ expected drift direction.
- Compatibility note required by AoC_03 (α) and any later SM/QFT appendices.

Primary Dependencies

- AoC_00_Atlas_Preface (epistemic tags + Units Firewall discipline)
- Appx_04_On_Zero_PF (ζ bounds/regularization meaning)
- Appx_06_On_Chance_PF (ξ fluctuation/selection; stochastic contribution to flow)
- Appx_11_On_Gauge_Structure_STUB (what it means for gauge structure to persist across scale)

Primary Falsifier

PF cannot produce a predictive constraint on running (beyond restating standard RG), or it violates the Units Firewall by allowing survivor invariants to drift; in that case, “survivability bookkeeping” collapses into post-hoc interpretation.

Support

Appx_17_EFT_RG_Support/

(mapping drafts, flow-family candidates, counterexample audit, discriminator sheet, rewrite targets, HoR packaging plan)



Start here if:

you’re reconciling “running couplings” with constants-as-survivors, auditing circularity in QFT explanations, or building a discriminator that shows whether PF actually constrains RG flow rather than merely renaming it.